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Dictionary & Sets

- **Dictionary**

Dictionary is a collection of keys-value pairs

Syntax:

```
a = {"Key": "Value",
     "Malip": "python",
     "marks": "100"
     "list": [1,2,9]
    }
a["key"]  # print value
a[list]   # print [1,2,9]
```

- **Properties of Python Dictionaries**
  1. It is unordered
  2. It is mutable
  3. It is indexed
  4. cannot contain duplicate keys

- **Dictionary Methods**

```
a = {"name": "Malip"
     "from": "Bengaluru"
     "marks": "100"
}
```

1. a.items() : Returns a list of (key, value) tuples.
2. a.keys() : Returns a list containing dictionary keys.
3. a.update({"friends":}) : update the dictionary with supplied key-value pairs.

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a.get ("name") : Returns the value of the specified keys (and values is returned eg. "Malup" is returned here)

Sets in Python

Set is a collection of non-repetitive elements.
Such as

```
S = set ()        # no repetition allowed
S.add (1)         //
S.add (2)         # our set = {1,2}
```

Properties of Sets

1. Sets are unordered.
2. Sets are unindexed.
3. There is no way to change items in sets.
4. Sets cannot contain duplicate values.

Sets Methods

Consider the following sets:

S = {1, 8, 2, 3}

- len(S) : Return 4, the length of the set
- S.remove(8) : Update the set S and remove 8 from it.
- S.pop() : Remove an arbitary element from set and return the element removed.
- S.clear() : empties the set S.
- S.union({8,11}) : Return a new set with all items from both sets {1,8,2,3,11}
- S.intersection({8,11}) : Return a set which only contains only items in both sets.

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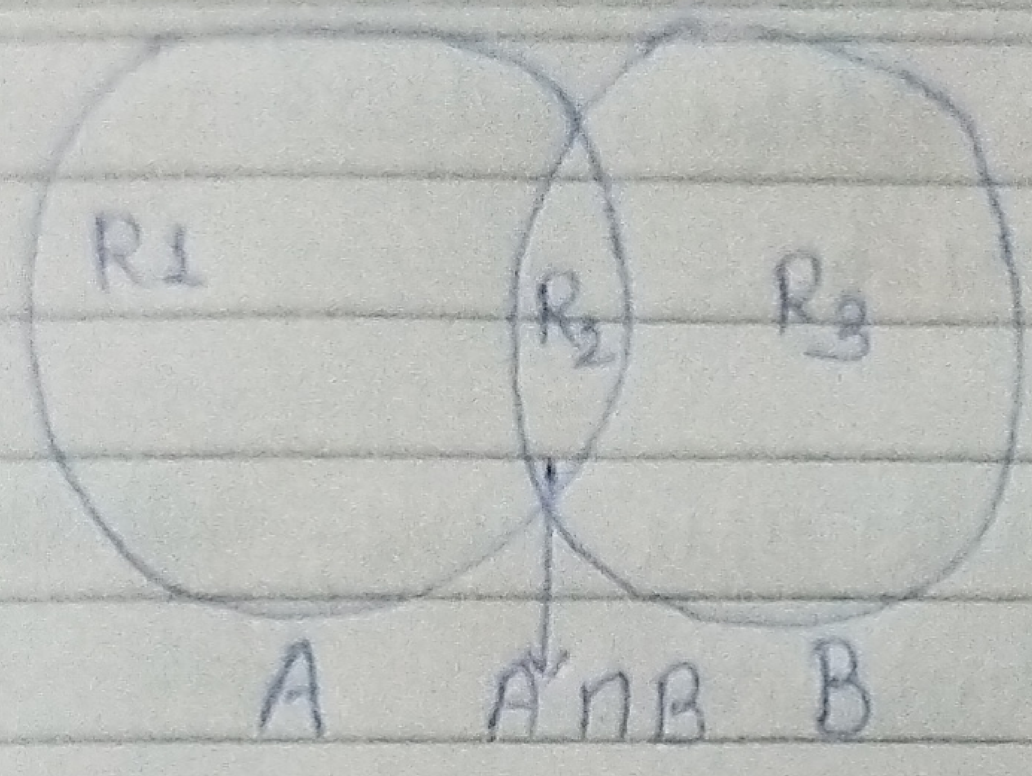

$R_2 = A \cap B$

$R_1 + R_2 + R_3 = A \cup B$

$R_1 + R_3 = A \Delta B$

$R_1 = A - B$

$R_3 = B - A$

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PRACTICE-SET

1) Write a Program to create dictionary of Hindi word with values as their English translation. Provide user with an option to look it up.

2) Write a Program to input 8 numbers from the user and display all the unique numbers once.

3) Can we have a set with 18 (int) and "18" (str) value in it.

4) What will be the length of the following set S:

```
S = Set()
s.add(20)
s.add(20.0)
s.add("20")
```

5) S = {} what is the type of S.

6) Create an empty dictionary. Allow 4 friends to enter their favourite language as value and use key as their names. Assume that their names are unique.

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7) If the name of 2 friends are same; what will happen to the program in Problem 6.

8) If the language of 2 friends are same; what will happen to the program in problem 6.

9) Can you change the values inside a list which is contained in set S?

$S = \{8, 7, 12, "Harry", [1, 2]\}$